

GreenTrail Outdoors

Promotion Strategy: Final Report (Revised)

Historical promotion analysis and Test & Learn framework

Executive summary

GreenTrail wants in-store promotions that are commercially effective and consistent with its sustainability ethos. We analysed 25 store-promotion observations (three promotion types, two store formats, April to August 2023) and used them to design a controlled Test & Learn programme.

What happened. Average weekly sales were \$36,800 before, \$49,440 during and \$38,280 after promotion. Average store-level sales uplift was 35.23% during and 4.59% after, so only about 13% of the lift remained once promotions ended. Visits rose 25.67% during and 7.00% after, and stores with bigger visit gains saw bigger sales gains (correlation 0.85).

What differs. Special Event kept the most lift afterwards (+9.94%) and none of its stores ended below baseline. BOGO was weakest in both store formats and ended below baseline (-1.94%). Urban stores responded more than suburban. **New in this revision:** Special Event ran in more urban stores (5 of 8) than Discount (4 of 9), so we adjusted for store mix. After adjustment, Discount (38.1%) and Special Event (37.5%) are effectively tied during the promotion; Special Event's edge is in what lasts afterwards (+9.4% vs +6.4%).

What we recommend. Treat these as hypotheses, not proof. Test a sustainability-themed Special Event and a targeted Discount on durable products, with BOGO as a benchmark, against matched control stores for 4 weeks plus a 2-week follow-up. BOGO is not recommended as a lead mechanic: it was the weakest and least lasting in the data, and it rewards buying more units than needed, which conflicts with GreenTrail's positioning.

1. Business problem

GreenTrail's promotions are not achieving the desired impact and must also respect the company's sustainability values. The work answers five questions: what happened in past promotions, what data is missing, how to build a fair control group, how long to test, and what a data-driven, sustainable strategy looks like. The historical file supports descriptive analysis only; it does not establish causal impact.

2. Data and methodology

Item	Description
Observations	25 store-promotions (one per store); 14 variables
Promotion types	Buy-One-Get-One (BOGO), Discount, Special Event
Store types	Urban, Suburban
Timing	Promotions started between April and August 2023; 22 ran 15 days and 3 ran 14 days
Metrics	Weekly sales and average daily visits, before, during and after each promotion

Data quality checks (01_GreenTrail_DataCleaning.sql): 0 missing values, 0 duplicate store IDs, 0 negative sales or visit values, 0 promotions ending before they start, and 0 stores with sales below baseline during their promotion. Each of the 25 locations appears once.

Uplift convention. Store-level percentage uplift is calculated first, then averaged for group comparisons: sales uplift during = (during - before) / before x 100, and likewise for after and for visits. Uplift decay = uplift during - uplift after. As a cross-check, pooled uplift (ratio of total sales) is 34.35% during and 4.02% after, close to the store-level averages of 35.23% and 4.59%, so the conclusions do not depend on the convention.

3. Historical analysis

3.1 Overall sales and traffic

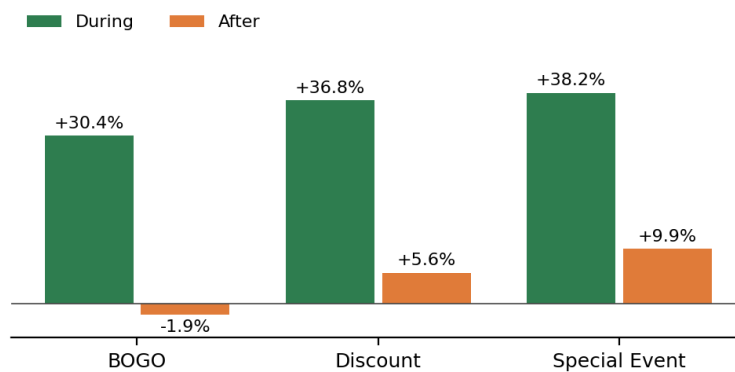
Metric	Before	During	After
Average weekly sales	\$36,800	\$49,440	\$38,280
Average daily visits	357.2	447.0	380.2
Average store-level sales uplift	-	+35.23%	+4.59%
Average store-level visit uplift	-	+25.67%	+7.00%

3.2 Promotion type and store type

Promotion	Stores	Sales during	Sales after	Visits during	Visits after
Buy-One-Get-One	8	+30.44%	-1.94%	+20.72%	+0.25%
Discount	9	+36.84%	+5.64%	+26.32%	+6.94%
Special Event	8	+38.20%	+9.94%	+29.89%	+13.83%

Store type	Stores	Sales during	Sales after	Visits during	Visits after
Suburban	12	+28.15%	+0.23%	+16.59%	+1.78%
Urban	13	+41.76%	+8.61%	+34.06%	+11.82%

Sales uplift by promotion type



Sales uplift by store type

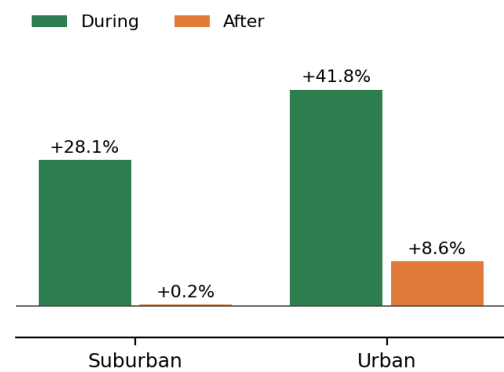


Figure 1. Average store-level sales uplift vs. the pre-promotion week.

3.3 Promotion x store type

Store	Promotion	N	Sales during	Sales after	Visits during	Visits after
Suburban	Buy-One-Get-One	4	+22.07%	-6.09%	+10.77%	-2.86%
Suburban	Discount	5	+29.45%	+1.34%	+18.40%	+3.62%
Suburban	Special Event	3	+34.07%	+6.81%	+21.32%	+4.90%
Urban	Buy-One-Get-One	4	+38.82%	+2.21%	+30.67%	+3.35%
Urban	Discount	4	+46.07%	+11.01%	+36.22%	+11.08%
Urban	Special Event	5	+40.68%	+11.82%	+35.03%	+19.19%

BOGO has the lowest uplift in both store types, during and after. Groups contain only 3 to 5 stores, so differences between Discount and Special Event within a store type are not reliable.

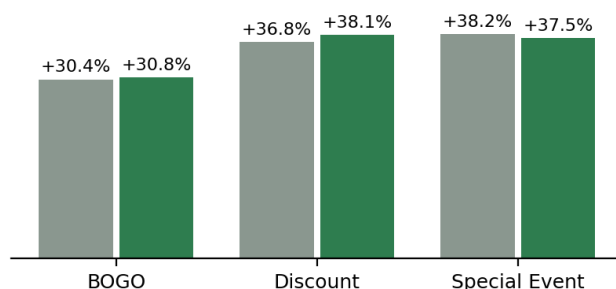
3.4 Store-mix check (new)

Promotions were not run in the same mix of stores: 5 of 8 Special Event stores were urban, compared with 4 of 9 Discount stores and 4 of 8 BOGO stores. Because urban stores respond more, part of Special Event's raw lead could come from where it ran. To check, we re-weighted each promotion's within-store-type results to the overall mix of the 25 stores (52% urban, 48% suburban).

Promotion	During: observed	During: mix-adjusted	After: observed	After: mix-adjusted
Buy-One-Get-One	+30.44%	+30.78%	-1.94%	-1.78%
Discount	+36.84%	+38.09%	+5.64%	+6.37%
Special Event	+38.20%	+37.51%	+9.94%	+9.42%

During promotion

As observed Mix-adjusted



After promotion

As observed Mix-adjusted

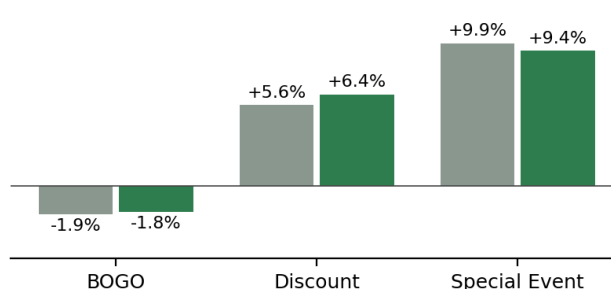


Figure 2. Observed vs. mix-adjusted sales uplift by promotion type.

Reading: BOGO stays clearly weakest. Discount and Special Event are effectively tied during the promotion once store mix is accounted for, while Special Event keeps the stronger result afterwards. This is still a small-sample adjustment, so it sharpens the hypotheses to test rather than settling them.

3.5 Store size and location

The correlation between store size and sales uplift during promotion is 0.62. Urban stores are also larger on average (1,704 vs 1,508), so size and store format overlap and cannot be separated here. Size is best treated as a segmentation and matching variable, not a cause. Location appears once per row, so location-level results are single-store values and are not interpreted as location effects.

3.6 Persistence and anomalies

- 7 of 25 stores ended below their pre-promotion sales: BOGO 4 of 8, Discount 3 of 9, Special Event 0 of 8.
- Store 1 (urban, Discount) had the highest sales uplift (66.7%); suburban Stores 17 and 7 had the lowest (4.8% and 7.7%). These are worth investigating for local factors.
- No store fell below baseline during its promotion. Promotion lengths were almost identical (14 to 15 days), so duration effects cannot be separated.
- All promotions ran April to August, so the pre/during/after comparison may include seasonal movement that a control group would remove.

4. Key findings

- Promotions are associated with a strong short-term response (35.23% sales uplift) that largely fades (4.59% afterwards); the post-period should be a success metric, not only the promotion period.
- Traffic moves with sales (visit uplift 25.67% during, 7.00% after; correlation with sales uplift 0.85). Promotions appear to buy footfall more than larger baskets: spend per visit was about \$14.72 before, \$15.80 during and \$14.38 after.
- BOGO is the weakest promotion in both store formats and the only one with negative post-promotion sales uplift overall.
- Special Event shows the best persistence; Discount is equally strong during the promotion once store mix is accounted for.
- Urban stores respond more than suburban stores; store format, size and promotion type overlap and need matching in the test.
- These are observational patterns from a small sample and cannot be read as causal effects.

5. Additional data requirements

Data	Why it matters
Non-promoted comparison stores	Provides a control group and a counterfactual.
Transaction-level behaviour	Measures transactions, units, basket value and conversion.
Customer segments and store characteristics	Shows who responds and improves matching of test and control stores.
Promotion cost and discount depth	Separates sales uplift from profit uplift.
Regional economic data, weather, local events, competitor activity	Separates outside effects from promotion effects.
Sustainability KPIs	Measures sustainability alongside commercial results.

6. Control strategy

Create treatment and control stores before the test, and decide the matching before seeing results. Match or stratify on store type, store size, baseline sales, baseline traffic and market characteristics. Control stores must not receive the tested intervention during the test window.

Incremental impact is estimated with a Difference-in-Differences comparison: **(test after - test before) - (control after - control before)**. A plain before/after comparison would credit the promotion with everything that changed, including seasonality.

Illustrative example (invented numbers)	Before	During test	Change
Test stores, weekly sales per store	\$40,000	\$50,000	+25%
Control stores, weekly sales per store	\$40,000	\$43,200	+8%
Incremental effect of the promotion			+17 points (about \$6,800 per store per week)

The example uses invented numbers to show the method; it is not a result. A before/after view would claim +25%; the control group shows that +8% would have happened anyway.

7. Test duration and sample size

Duration. Working design: 6 weeks, made up of a 4-week intervention (several full weekly shopping cycles, versus the 2-week promotions seen historically) and a 2-week post-period to measure persistence. The window and success thresholds are fixed before launch and not extended because of weak results; holidays and local events are tracked. The exact length should be validated against seasonality and the business calendar.

Number of stores (new). A rough planning calculation, using the historical spread of store-level sales uplift (standard deviation 14.6 percentage points) as a proxy, a two-sided 5% significance level and 80% power, gives the stores needed *per group* to detect a given difference in uplift between a test group and control:

Difference to detect (uplift points)	Stores needed per group
10 points	34
15 points	15
20 points	9

With three promotion arms plus a control group, detecting a 15-point difference needs roughly 60 stores in total, and smaller differences need more. Matching stores and using weekly data in a Difference-in-Differences design should reduce the variation and the number needed, so this is a planning estimate to be confirmed with real store data. If GreenTrail's store base cannot support four groups, test Special Event and Discount against control first and run BOGO as a benchmark in a later wave.

8. Strategic recommendation and sustainability fit

Sustainability is a requirement, not an add-on. A promotion should be judged on whether it lifts sales in a way that fits GreenTrail's brand: encouraging useful, durable purchases and participation rather than unnecessary consumption. The dataset contains no sustainability measures, so the points below are hypotheses to test.

Promotion	What the data shows	Sustainability fit (hypothesis)	Role in test
Special Event	Best persistence (+9.94% after, 0 of 8 stores below baseline); tied with Discount during the promotion after store-mix adjustment	Events can be built around repair clinics, gear swaps, trail clean-ups and guided outings, rewarding participation instead of volume	Lead candidate (Arm A)
Discount	Strong during (+36.84%), moderate after (+5.64%)	Fits better if targeted at durable, repairable or sustainably made products, not blanket price cuts; margin impact must be tracked	Second candidate (Arm B)
BOGO	Weakest in both store types; -1.94% after; 4 of 8 stores below baseline	Rewards buying extra units, which can conflict with a sustainable-consumption message	Benchmark only (Arm C)

Sustainability KPIs to agree with GreenTrail's sustainability team: sustainable-product share of sales, participation in repair, refill or swap activities, returns and waste, packaging or resource impact, and repeat visits after the promotion.

9. Proposed Test & Learn framework

- Define hypotheses and success thresholds before launch.
- Arms: A) sustainability-themed Special Event, B) targeted Discount on durable products, C) BOGO benchmark, D) matched control stores with no new promotion.
- Assign matched treatment and control stores before results are observed.
- Measure incremental sales (and gross profit where available), transactions, basket value, visits, post-period performance and sustainability KPIs.
- Track weather, holidays, local events, competitor activity and stock-outs.
- Evaluate with Difference-in-Differences against a predefined statistical approach, document the result and use it to design the next test.

Established from history (observation)	Only a controlled test can establish
Promotions are followed by higher sales and visits	How much of the lift is truly incremental
BOGO had the weakest and least lasting results	Whether BOGO is worse because of the mechanic or the stores it ran in
Special Event kept the most lift afterwards	Whether Special Event causes the persistence, and whether it holds in all store types
Urban stores responded more than suburban	Whether the promotion should differ by store format
Sustainability data is not in the file	Whether sustainability-themed promotions protect the brand and commercial results

10. Limitations

- Small sample (25 observations; 3 to 5 stores per promotion and store-type group).
- Observational data with no clean control group, so no causal identification.
- Promotion type, store type and store size overlap (for example, Special Event ran in more urban stores).
- All promotions ran in the same April to August window; seasonality, weather, events and competitors are not in the data.
- No transaction, customer, promotion-cost or sustainability data; sales and visits are aggregated, so conversion and profit cannot be calculated.
- Each location appears once, and promotion lengths are almost identical (14 to 15 days).

Appendix: SQL (02_Greentrail_Analysis_v2.sql)

Data cleaning is in 01_GreenTrail_DataCleaning.sql; extra checks used in the presentation are in 03_GreenTrail_Storytelling.sql. The analysis script is reproduced in full below with long lines wrapped.

```
USE mastercard;

-- GreenTrail promotion analysis (v2)
-- Convention: store-level % uplift is calculated first, then averaged for group comparisons.
-- Changes from v1: 2-decimal rounding on store-level uplifts, pooled-uplift check, store-mix check,
-- mix-adjusted promotion comparison, correlations in SQL, and a location data check.

-- 1. Overall Sales Performance
SELECT
ROUND(AVG(weekly_sales_before_promotion), 2) AS avg_sales_before,
ROUND(AVG(weekly_sales_during_promotion), 2) AS avg_sales_during,
ROUND(AVG(weekly_sales_after_promotion), 2) AS avg_sales_after
FROM greentrail_store_data;

-- 2. Overall Customer Traffic
SELECT
ROUND(AVG(average_daily_visits_before_promotion), 2) AS avg_visits_before,
ROUND(AVG(average_daily_visits_during_promotion), 2) AS avg_visits_during,
ROUND(AVG(average_daily_visits_after_promotion), 2) AS avg_visits_after
FROM greentrail_store_data;

-- 3. Sales uplift per store (2 decimals; v1 rounded to whole numbers)
SELECT store_id, weekly_sales_before_promotion, weekly_sales_during_promotion, weekly_sales_after_promotion,
ROUND((weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100, 2) AS
sales_uplift_during_pct,
ROUND((weekly_sales_after_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100, 2) AS
sales_uplift_post_pct
FROM greentrail_store_data
ORDER BY store_id;

-- 4. Average store-level uplift vs pooled uplift (ratio of averages), to show the convention matters little
SELECT
ROUND(AVG((weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2)
AS avg_store_uplift_during_pct,
ROUND((SUM(weekly_sales_during_promotion) - SUM(weekly_sales_before_promotion)) / SUM(weekly_sales_before_promotion) *
100, 2) AS pooled_uplift_during_pct,
ROUND(AVG((weekly_sales_after_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2) AS
avg_store_uplift_post_pct,
ROUND((SUM(weekly_sales_after_promotion) - SUM(weekly_sales_before_promotion)) / SUM(weekly_sales_before_promotion) *
100, 2) AS pooled_uplift_post_pct
FROM greentrail_store_data;

-- 5. Visit uplift (average store-level)
SELECT
ROUND(AVG((average_daily_visits_during_promotion - average_daily_visits_before_promotion) /
average_daily_visits_before_promotion * 100), 2) AS avg_visit_uplift_during_pct,
ROUND(AVG((average_daily_visits_after_promotion - average_daily_visits_before_promotion) /
average_daily_visits_before_promotion * 100), 2) AS avg_visit_uplift_post_pct
FROM greentrail_store_data;

-- 6. Promotion type analysis
SELECT type_of_promotion, COUNT(*) AS no_of_stores,
ROUND(AVG((weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2)
AS sales_uplift_during_pct,
ROUND(AVG((weekly_sales_after_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2)
AS sales_uplift_post_pct,
ROUND(AVG((average_daily_visits_during_promotion - average_daily_visits_before_promotion) /
average_daily_visits_before_promotion * 100), 2) AS visit_uplift_during_pct,
ROUND(AVG((average_daily_visits_after_promotion - average_daily_visits_before_promotion) /
average_daily_visits_before_promotion * 100), 2) AS visit_uplift_post_pct
FROM greentrail_store_data
GROUP BY type_of_promotion
ORDER BY no_of_stores DESC;

-- 7. Store type analysis
SELECT store_type, COUNT(*) AS no_of_stores,
ROUND(AVG(weekly_sales_before_promotion), 2) AS avg_sales_before,
ROUND(AVG(weekly_sales_during_promotion), 2) AS avg_sales_during,
ROUND(AVG(weekly_sales_after_promotion), 2) AS avg_sales_after,
ROUND(AVG((weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2)
AS sales_uplift_during_pct,
ROUND(AVG((weekly_sales_after_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2)
AS sales_uplift_post_pct,
ROUND(AVG((average_daily_visits_during_promotion - average_daily_visits_before_promotion) /
average_daily_visits_before_promotion * 100), 2) AS visit_uplift_during_pct,
ROUND(AVG((average_daily_visits_after_promotion - average_daily_visits_before_promotion) /
average_daily_visits_before_promotion * 100), 2) AS visit_uplift_post_pct
FROM greentrail_store_data
```

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GROUP BY store_type;

-- 8. Promotion x store type (which promotion worked for which store type)
SELECT store_type, type_of_promotion, COUNT(*) AS no_of_stores,
ROUND(AVG((weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2)
AS sales_uplift_during_pct,
ROUND(AVG((weekly_sales_after_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2)
AS sales_uplift_post_pct,
ROUND(AVG((average_daily_visits_during_promotion - average_daily_visits_before_promotion) /
average_daily_visits_before_promotion * 100), 2) AS visit_uplift_during_pct,
ROUND(AVG((average_daily_visits_after_promotion - average_daily_visits_before_promotion) /
average_daily_visits_before_promotion * 100), 2) AS visit_uplift_post_pct
FROM greentrail_store_data
GROUP BY store_type, type_of_promotion
ORDER BY store_type, type_of_promotion;

-- 9. NEW: store-type mix inside each promotion (is store type mixed up with promotion type?)
SELECT type_of_promotion, COUNT(*) AS no_of_stores,
SUM(CASE WHEN store_type = 'Urban' THEN 1 ELSE 0 END) AS urban_stores,
ROUND(SUM(CASE WHEN store_type = 'Urban' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 0) AS urban_share_pct
FROM greentrail_store_data
GROUP BY type_of_promotion
ORDER BY type_of_promotion;

-- 10. NEW: mix-adjusted promotion comparison
-- Each promotion's uplift within a store type is re-weighted to the overall store-type mix of all 25 stores,
-- so promotions are compared as if each had run in the same mix of urban and suburban stores.
WITH cell AS (
SELECT store_type, type_of_promotion,
AVG((weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100) AS
during_uplift,
AVG((weekly_sales_after_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100) AS
post_uplift
FROM greentrail_store_data
GROUP BY store_type, type_of_promotion
),
weights AS (
SELECT store_type, COUNT(*) * 1.0 / (SELECT COUNT(*) FROM greentrail_store_data) AS w
FROM greentrail_store_data
GROUP BY store_type
)
SELECT c.type_of_promotion,
ROUND(SUM(c.during_uplift * w.w), 2) AS mix_adjusted_during_uplift_pct,
ROUND(SUM(c.post_uplift * w.w), 2) AS mix_adjusted_post_uplift_pct
FROM cell c
JOIN weights w ON c.store_type = w.store_type
GROUP BY c.type_of_promotion
ORDER BY c.type_of_promotion;

-- 11. Store size: store-level view and correlation with sales uplift
SELECT store_id, store_size, store_type,
ROUND((weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100, 2) AS
sales_uplift_during_pct
FROM greentrail_store_data
ORDER BY store_size;

-- MySQL has no CORR(), so Pearson correlation is built from AVG and STDDEV_POP
SELECT ROUND((AVG(x * y) - AVG(x) * AVG(y)) / (STDDEV_POP(x) * STDDEV_POP(y)), 3) AS corr_store_size_vs_sales_uplift
FROM (
SELECT store_size AS x,
(weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100 AS y
FROM greentrail_store_data
) t;

SELECT ROUND((AVG(x * y) - AVG(x) * AVG(y)) / (STDDEV_POP(x) * STDDEV_POP(y)), 3) AS corr_visit_uplift_vs_sales_uplift
FROM (
SELECT (average_daily_visits_during_promotion - average_daily_visits_before_promotion) /
average_daily_visits_before_promotion * 100 AS x,
(weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100 AS y
FROM greentrail_store_data
) t;

-- 12. Location data check (replaces the v1 location analysis)
-- Every location appears once, so location-level averages are single-store values, not reliable location effects.
SELECT COUNT(*) AS total_rows, COUNT(DISTINCT location) AS distinct_locations
FROM greentrail_store_data;

-- 13. Promotion response persistence / decay
SELECT type_of_promotion,
ROUND(AVG((weekly_sales_during_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2)
AS during_sales_uplift_pct,
ROUND(AVG((weekly_sales_after_promotion - weekly_sales_before_promotion) / weekly_sales_before_promotion * 100), 2)
AS post_sales_uplift_pct,
ROUND(AVG((weekly_sales_during_promotion - weekly_sales_after_promotion) / weekly_sales_before_promotion * 100), 2) AS
uplift_decay_pct
FROM greentrail_store_data

```

GROUP BY type_of_promotion;